

Merlin — Technical Appendix

Adaptiv Systems · Investor / due-diligence package

Version: 1.0 · June 2026 · Confidential

Production: merlin.adaptiv.systems · Mobile: mobile.adaptiv.systems

Document map

Section	Topic
1–8	Architecture (stack, tenancy, data model, agents, surfaces)
9	Engineering process & CI
10	Test strategy
11	Security & subprocessors
12	Scalability (~100 tenants)
13	Known gaps (honest)
14	Governance policies
15	90-day engineering priorities

Regenerate: `python3 scripts/generate-technical-appendix.py` → Merlin-Technical-Appendix.docx + .pdf

1. Executive summary

Merlin is Adaptiv Systems' operating layer for physical spaces — multi-tenant B2B SaaS where AI agents observe building signals and work alongside facility managers and contractors. The stack:

- **Frontend:** React 18 SPA (Vite), four surfaces (customer, worker, mobile field app, platform admin)
- **Backend:** ~113 TypeScript serverless functions on Vercel + 15 crons
- **Data:** Supabase Postgres with Row-Level Security (RLS), Auth, Storage, Realtime
- **AI:** Anthropic Claude (streaming chat + agent ticks)

Baseline (June 2026): ~200K LOC · 259 forward-only migrations · 1,709 commits · 10 CI jobs · 21 Vitest suites · 3 Playwright E2E journeys · Node 22.

Merlin passes Series-A technical due diligence on schema, multi-tenancy, and CI. Remaining investment is **governance packaging** (this document + policies), **2 more E2E journeys**, and **ops proof** (backup drill, load test) — not an architectural rewrite.

2. System stack

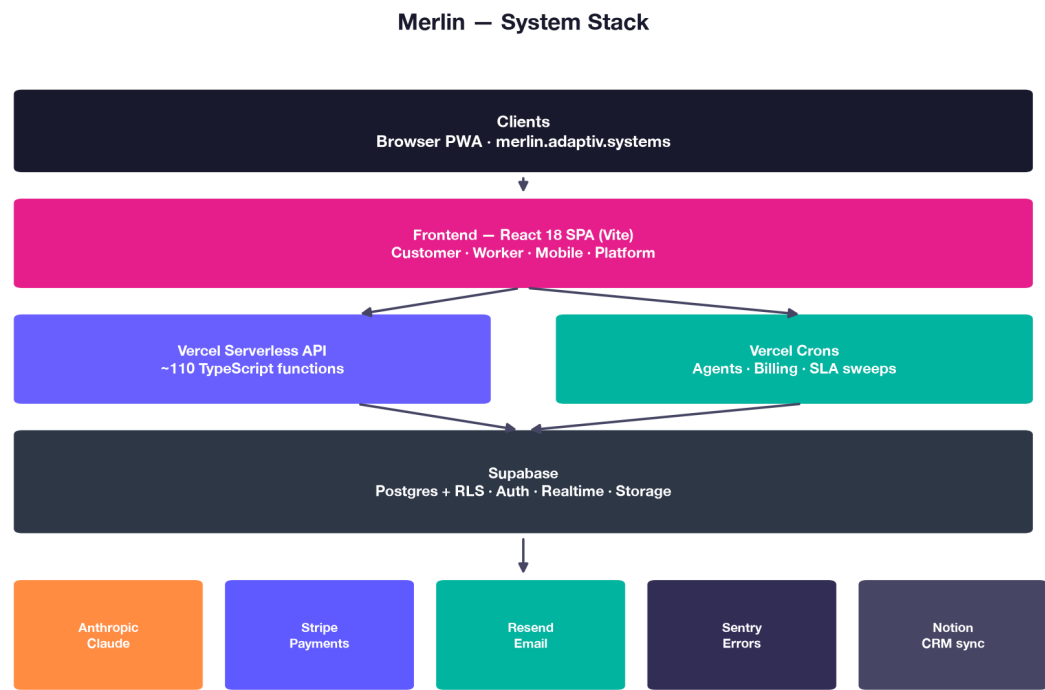


Figure 1: System stack

Layer	Technology
Frontend	React 18 · Vite · lazy routes · i18n EN/FR/DE (+ ES/PT mobile)
API	Vercel Serverless Functions (100% TypeScript)
Database	Supabase Postgres + RLS · Auth · Realtime · Storage
AI	Anthropic Claude
Payments	Stripe LIVE (subscriptions, agent add-ons, Customer Portal)
Email	Resend + Supabase Auth SMTP (SPF/DKIM/DMARC on adaptiv.systems)
Monitoring	Sentry (all API/cron handlers + 22 frontend data-layer modules)
Health	/api/health — cold start + DB round-trip

Hosting: **Vercel** + **Supabase** (single region). Optional AWS migration documented but not required for current scale.

3. Building operations model

Building Operations — Commercial Hierarchy

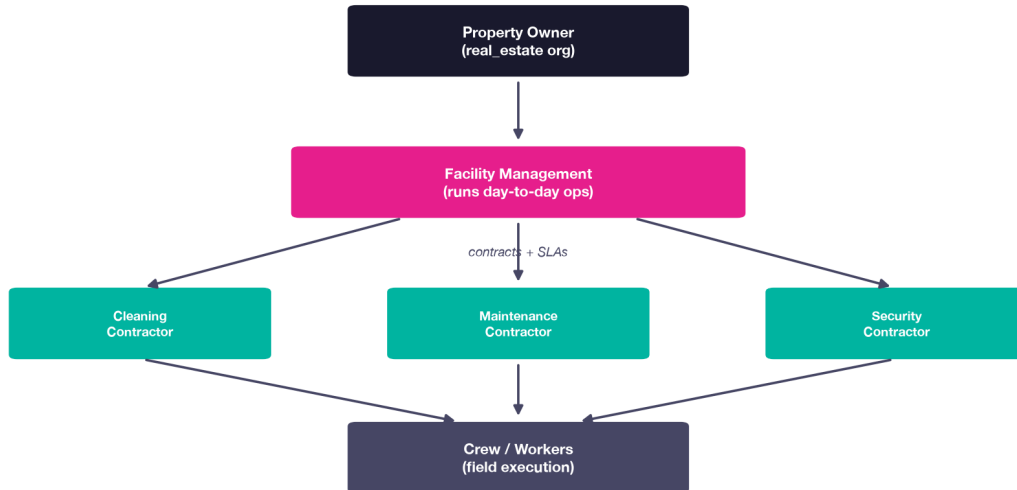


Figure 2: Org hierarchy

Organizations have a **kind**: `real_estate` (owners/FMs), `contractor` (service firms), or `adaptiv` (platform). Locations carry `owner_org_id` and `manager_org_id`. **Contracts** link facility managers and contractors with SLAs — first-class entities, not afterthoughts.

Eight orgs run in production today (Meridian multi-building flagship, FEB, IMF demo, SparkleCo + contractor ecosystem, Adaptiv platform).

4. Core data model

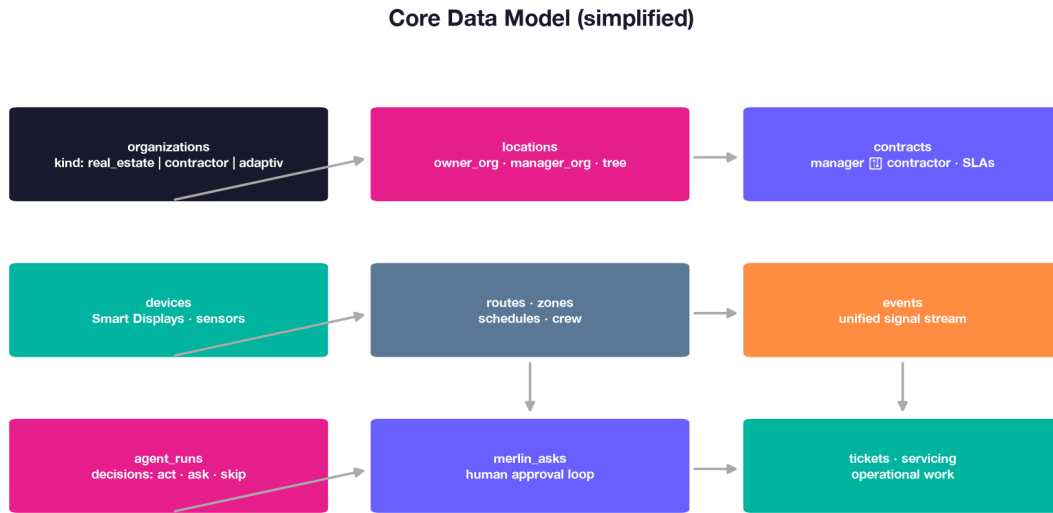


Figure 3: Data model

Center: **organizations** → **location tree** → **contracts** → **devices/events** → **agent_runs** → **merlin_asks** / **tickets**. Demo orgs use **replay mode** (capture once, replay without Anthropic spend).

5. Multi-tenancy & security

Multi-Tenancy & Row-Level Security

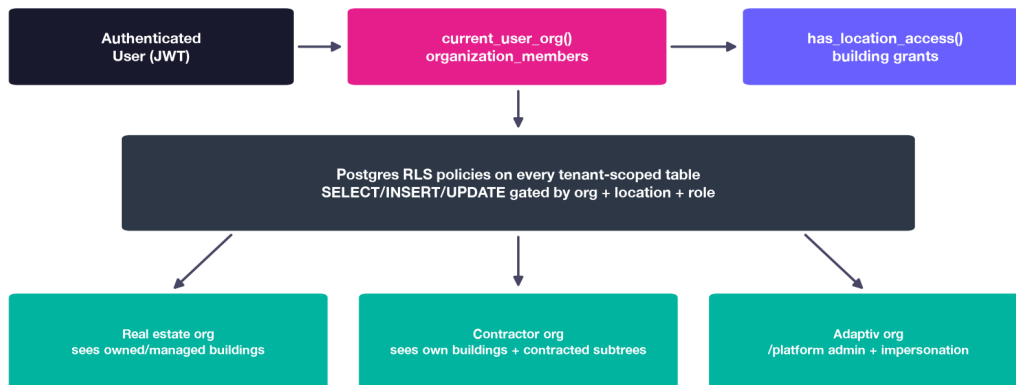


Figure 4: RLS

Every tenant-scoped table has RLS. Helpers (`current_user_org()`, `has_location_access()`, contractor contract paths) centralize policy logic across 259 migrations.

June 2026 hardening (migs 257–259): trigger functions no longer client-callable; anon stripped from 37 write RPCs; 33 internal DEFINER functions → `service_role` only.

CI: cross-tenant leak regression suite (read-only prod as Lisa@SparkleCo) · RLS scope linter on client reads · mobile worker RPC party guard.

6. Events pipeline

Events Pipeline — Signal to Action

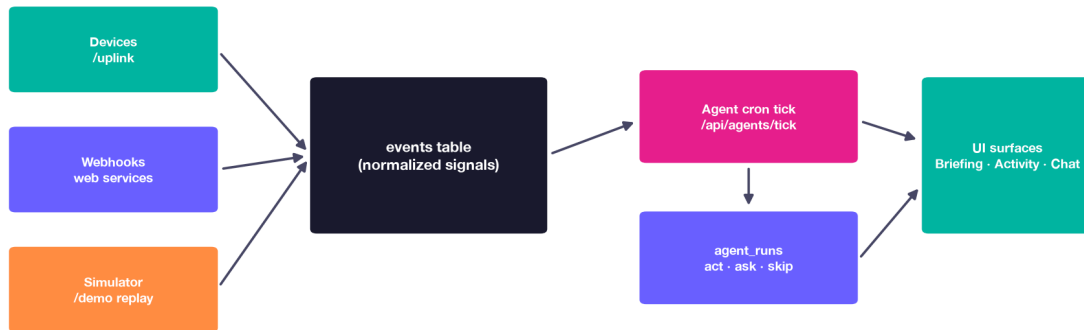


Figure 5: Events pipeline

Signals normalize to **events** → agent cron tick → **agent_runs** (act / ask / skip) → UI (briefing, activity, chat). Human approval via **merlin_asks** before autonomous side effects.

7. Application surfaces

Application Shells & Routes

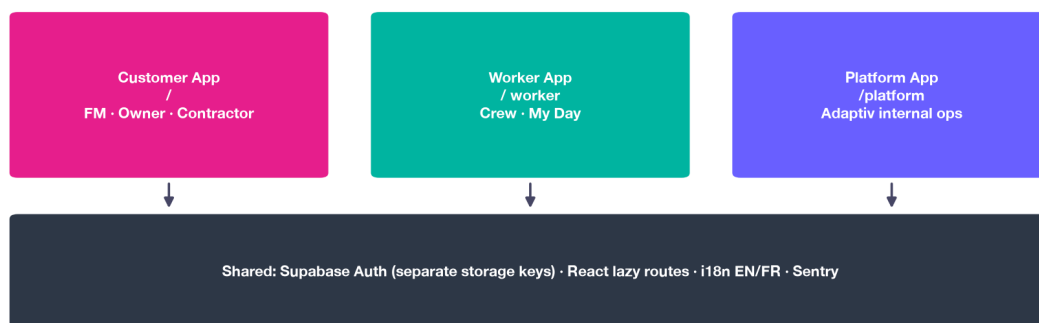


Figure 6: App shells

Surface	Route / host	Users
Customer	/	FM, owners, contractor managers
Worker	/worker	Desktop crew / My Day
Merlin Field	mobile.adaptiv.systems	Phone-first field workers (PWA)
Platform	/platform	Adaptiv internal ops

Separate Supabase auth storage keys per shell. Mobile worker writes use **party-guarded RPCs** (mig 253), not bare INSERT policies.

8. Customer UX — five pillars

Customer UX — Five Operational Pillars

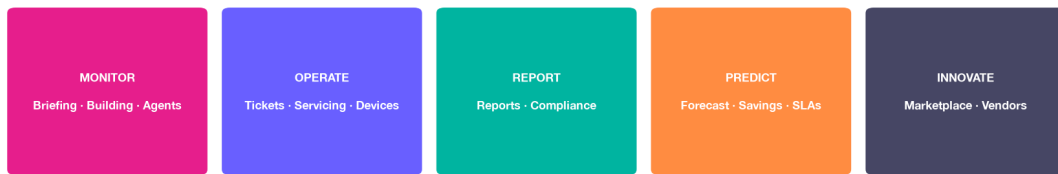


Figure 7: Pillars

MONITOR · OPERATE · REPORT · PREDICT · INNOVATE — navigation maps to briefing, tickets, compliance, forecasting, marketplace.

9. Engineering process

9.1 Branch & deploy

- **main** → production (Vercel auto-deploy after CI)
- PR required; template at `.github/pull_request_template.md`
- Husky pre-commit: Prettier + ESLint on staged files
- Dependabot: npm minor/patch

9.2 CI pipeline (10 jobs)

Job	Blocking	Purpose
build	✓	Production bundle; fails on Rollup circular-chunk warnings
typecheck (api)	✓	tsc --noEmit — 113/113 API files
typecheck (frontend)	✓	@ts-check opt-in modules (27 today)
lint-i18n	✓	Missing translation keys
lint-rls	✓	Org scope on client table reads
lint-js	✓	no-undef, rules-of-hooks = error
test	✓	Vitest — leak, money-path, smoke, units
e2e	✓	Playwright — 3 hermetic journeys
lint-deadcode	informational	Knip — ratchet planned
audit (npm)	informational	High/critical advisories

Docs-only PRs skip CI (docs/**, **/*.md).

9.3 Code structure (team onboarding)

Monolith decomposition sprint (June 2026):

Module	Before	After
MobileApp.jsx	2,136	~786 (shell) ✓
Admin.jsx	5,088	~190 (shell) ✓
Dashboard.jsx	2,839	~384 (shell) ✓
Chat.jsx	2,358	~1,565 🟡 in progress

10. Test strategy

10.1 Vitest (21 files)

Category	Examples
Tenant isolation	cross-tenant-leak.test.js (prod read-only)
Money paths	money-paths, contract-costs-guards, merlin-config-tenant-write
Smoke	page-smoke (~33 routed leaves)
Mobile	worker-rpc-guard, mobile-surface
Units	pagination, personas, autonomy-gate, stripe-config, push-dispatch, calls-approve
Drift guards	route-map, engines-node

Prod write safety: ALLOW_DESTRUCTIVE gate in test setup.

10.2 Playwright E2E (3 hermetic specs)

All use stub Supabase host + fixture intercepts — no prod secrets, deterministic CI.

Spec	Journey
e2e/mobile-happy-path.spec.js	Worker: mark task done → raise ticket with photo
e2e/login.spec.js	Password sign-in → signed-in surface
e2e/ask.spec.js	Worker Ask Merlin tab → reply in thread

Open: desktop approve-flow, contractor login (target 5 total journeys).

10.3 What tests do not cover yet

- HTTP handler integration (Stripe webhook signature, auth middleware)
- Load / concurrency (k6 planned on sandbox restore)
- Full Admin UI (excluded from page-smoke — auth + providers)

11. Security & subprocessors

11.1 Policies (lightweight)

Located in policies/:

1. Information Security Policy
2. Access Control Policy
3. Incident Response Policy
4. Change Management Policy

5. Data Retention & Deletion Policy

11.2 Subprocessors

Register: subprocessors/README.md — Supabase, Vercel, Stripe, Resend, Anthropic, Sentry. Collect DPA/SOC 2 PDFs per vendor folder before enterprise deals.

11.3 Open security backlog

Tracked in ../operations/security-deferred.md. Not SOC 2 certified — inherit from vendors (Supabase Team + Vercel Enterprise when required).

12. Scalability (~100 tenants)

See ../operations/100-tenants-readiness.md.

Done		Open	
Pagination past PostgREST 1K cap		Realtime org-scoping on subscriptions	
Per-tenant write rate limiting (mig 066)	RLS	performance (pg_stat_statements)	audit
Composite indexes (mig 064)		Executed PITR drill (procedure exists)	
Cross-tenant leak CI suite		Load test (sandbox)	
RPC least-privilege (migs 257–259)		tenant_seed_starter provisioning RPC	

Architecture supports ~100 paying tenants on current Vercel + Supabase Pro stack; Supabase Team tier at first enterprise SOC 2 ask.

13. Known gaps (June 2026)

Honest list for due diligence — builds trust.

Gap	Severity	Target
Technical Appendix PDF + policies	Medium	This deliverable
2 more Playwright E2E journeys	Medium	~1 week
Sentry → Slack metric alerts	Medium	~half day
Backup PITR drill (dated record)	Medium	~30 min execute
Chat.jsx split completion	Low	~1 week
App.jsx >2K lines	Low	After Chat
Knip blocking in CI	Low	~1 day
GDPR export + hard delete	Medium (EU)	1–2 weeks when GTM requires
Subprocessor DPA folder complete	Medium	Collect PDFs per deal
@ts-check net (27 → 50+)	Low	Ongoing

14. 90-day engineering priorities

Aligned with ../operations/professionalization-roadmap.md:

Priority	Deliverable
 Done	Admin + Dashboard splits · 3 E2E · full Sentry wrap · RPC hardening
Now	Finish Chat split · desktop + contractor E2E
Next	Sentry alerts · backup drill doc · knip ratchet
Then	App.jsx split · load test · @ts-check expansion
Business	10+ paying tenants · enterprise pilot narrative

15. References

Doc	Path
Code audit	../operations/code-preparedness.md
Roadmap	../operations/ professionalization-roadmap.md
Series B checklist	../operations/series-b-readiness.md
Architecture canon	../architecture/
Backup runbook	../operations/runbooks/backup-restore.md

*Generated from Merlin repo docs/ · Diagrams in _diagram-assets/ · Regenerate with scripts/
generate-technical-appendix.py*